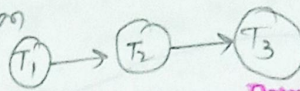


Assumption



Ques).

Page no: / / Date: / /

Page no: / /

oldest.

(100)

T_1

R(A)

W(C)

R(C)

(200)

T_2

R(B)

W(B)

(300)

T_3

R(B)

W(A)

youngest

	A	B	C
RTS	0	0	0
WTS	0	0	0

← Initially zero है सो

Soln).

	A	B	C
RTS	100 0	200 0	300 0
WTS	0	0	100

यहाँ पर T_2 Rollback हो रहा है बाकि सब Pass/success हो गया.

HD को हमको logically divide करेंगे

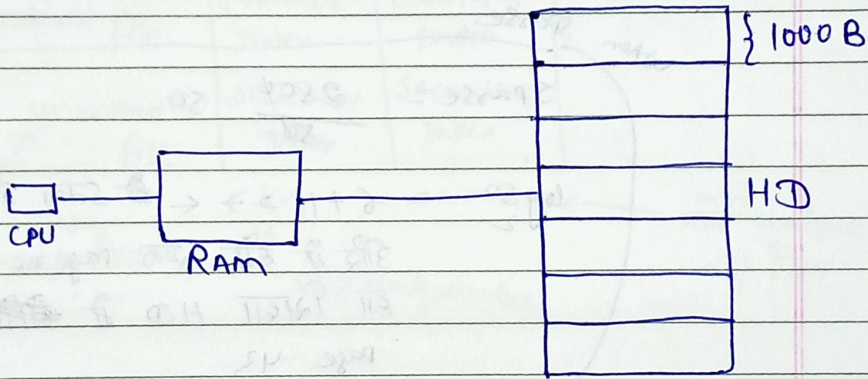
Indexing.

Like C, D E, f तब searching

Date: / / Page no:
 आसान है सोके।

Consider a Hard disk in which Block Size = 1000 Bytes each record is of size = 250 Bytes. If total no. of records are 10000 and the data entered in Hard disk w/o any order (unorder). What is avg. time complexity to Search a record from HD?

Soln.



$$\text{No. of records we can put in every block} = \frac{1000}{250} = 4$$

$$\text{No. of blocks required} = \frac{10000}{4} = 2500$$

Time Complexity in terms of I/O cost =

$$\text{Best Case} = 1$$

$$\text{Worst Case} = 2500 = (N)$$

$$\text{Avg Case} = \frac{2500}{2} = \left(\frac{N}{2}\right)$$

$$O(N).$$

Part-2: What is avg. time complexity to Search a record from Index table If index table entry = 20B (key + pointer)
10B 10B

Ans.

$$\text{Blocksize in HD} = \text{Block size in index}$$

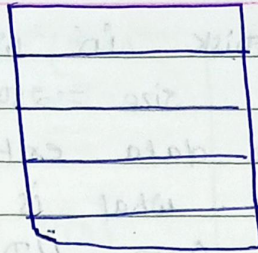
Sparse - इस ब्लॉक में एक entry.

↳ इस compressed होगा तब ही.

सब Data compressed

Dense → इस ब्लॉक index table में Data है।

Page no.



Index table

$$\text{Index table contains} = \frac{10000}{50} = 200$$

When sparse

$$\text{Sparse} = \frac{2000}{50} = 40$$

$\log_2 50 = 6 + 1 \Rightarrow \leftarrow$ इसका Search करना पड़ेगा और ये हमें एक page no. देगा जो कि हम सिधा H.D में ~~खोजेंगे~~ - ~~जाचेंगे~~ उस page पर

या हमें index table एक Newer Page देगा जो कि हमें पता चलेगा कि H.D के किस block में है।

When dense

$$\frac{10000}{50} = 200$$

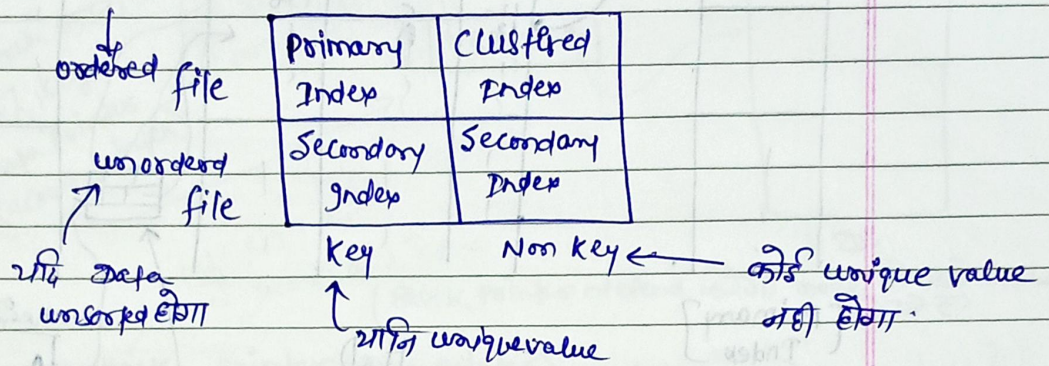
$$\text{Searching } \log_2 200 + 1 = 9.$$

table में एक से ज्यादा P.I. या C.I. नहीं बना सकते हैं।
 बनाने के लिए बना सकते हैं पर Database Admin
 ही करेगा। Date: / / Page no: _____

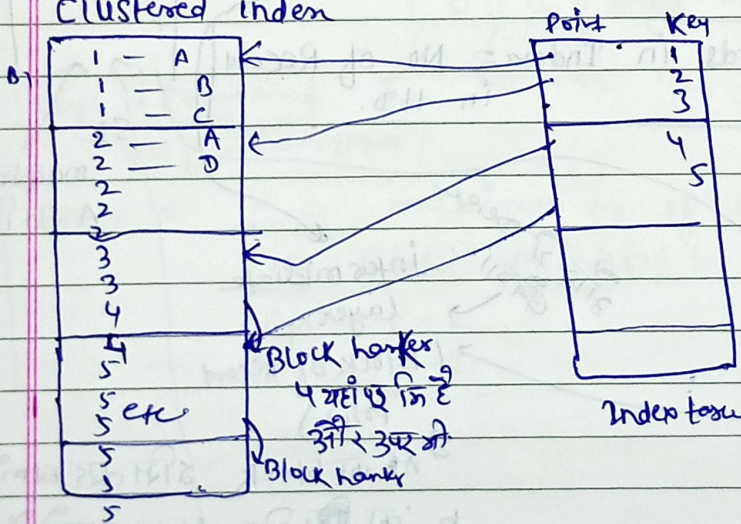
Type of indexes.

- 1) Primary Index ← sparse होता है mainly.
- 2) Clustered Index ← sparse always
- 3) Secondary Index

यदि data sorted होगा



Clustered index

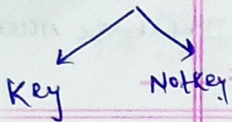


no. of block in index table

$$\text{time } \log_2 N + 1 + H$$

for header

Unordered data



Secondary Index

इसलिए Sorted में निवेश
ताकि Binary search
हो सके.

ये हमेशा Dense
होता।

(no Anchor)

Key	Pointer	id	name	PAN No.	Key	Pointer
1		1	A	40	23	
5		2	B	51	33	
9		3	A	62	40	
13		4	C	33	51	
		5	A	71	62	
		6	D	82	71	
		7	E	91	82	
		8	F	23	91	
		9	K	103	100	
		10	L	120	136	
		11	M	136		
		12				
		13				

[Primary Index]

HD

sensitive (error)

A	-
B	-
C	-
D	-

No. of Records in Index = No. of Records in HD.

वो A एका ही में है

time complexity

$$\log_2 N + 1 + 1 + \dots$$

intermediate
block 42
size of Pan

इसके बाद
आगे के

intermediate
layer

(Block of Record
pointer)

ये block इस तरह बनायेगे
b 'का बि बि block बनायेगे'
c " " " "
cfc.

SI

multiple
A में निशान

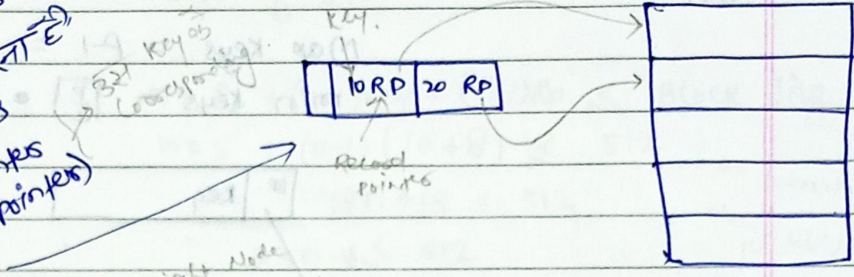
on the basis of
Non-Key.

सारे leaf node एक ही level पर होंगे

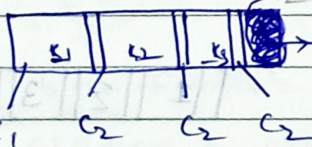
B-tree (Dynamic Multilevel Index)

	Root	Intermediate Node
max	P	P
min	2	$\lceil P/2 \rceil$

Keys $\rightarrow$ किस basis पर
search करता है
on basis of keys
Data pointers
(Record pointers)



Block pointer (Tree pointers)



Keys = Record pointers

दिया है $\log_2$ B tree

Order = P = maximum no. of children = 4

P = Block pointers

Keys = P - 1

RP = P - 1

a node can have

Ques)

Insert the following keys into B-tree, if order of B-tree = 4

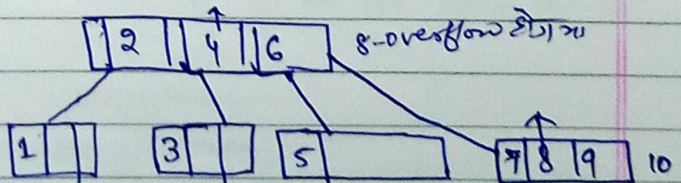
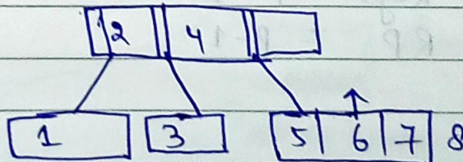
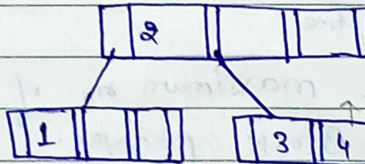
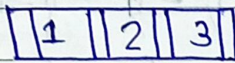
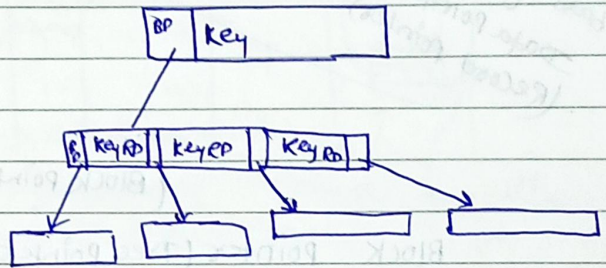
1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Soln.

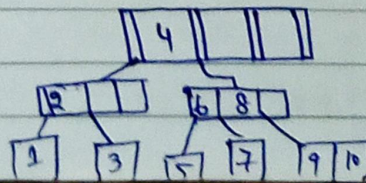
$$\text{Max Keys} = P-1 = 3$$

$$\text{min Keys} = \lceil \frac{P}{2} \rceil - 1 = 1$$

Follow the
Small element on
the left side of node
and large element on
the right side of
node.



height = 2
level = 3



Ans

Ans

min children = 2

order of B-tree is maximum no. of children possible

Ques) Consider a B-tree with Key Size = 10 bytes, block size 512 bytes, data pointer is of size 8 bytes and block pointer is 5 bytes. Find the order of B-tree?

Soln.

no. of block pointers in a node = n

size of one node = $n \times B_p + (n-1) \text{key} + (n-1)R_p$

$$n \times B_p + (n-1) \text{key} + (n-1)R_p \leq \text{Block Size}$$

$$n \times 5 + (n-1)(10+8) \leq 512$$

$$5n + 18n - 18 \leq 512$$

$$23n - 18 \leq 512$$

$$23n \leq 530$$

$$n \leq \frac{530}{23} = 23.04$$

23

no. of block pointers = 23

max. children = order = 23

min children = 2

min. children of internal node $\lceil \frac{23}{2} \rceil = 12$

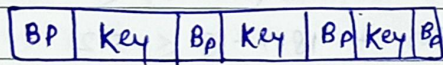
possible keys = order - 1

Maximum no. of Key, DP Pair in ^{leaf in} B+ is order of B+.

Question) Consider a B+ tree with key size = 10 bytes, block size = 512 bytes data pointer = 8 bytes and block pointer = 5 bytes. What is the order of leaf and non leaf node?

Solution.

Non leaf



Let block pointer = n

Size of node = $n \times BP + (n-1) \times \text{key} \leq \text{Block size}$

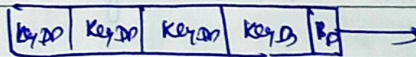
$$n \times 5 + (n-1) \times 10 \leq 512$$

$$n \leq \frac{512}{15} \approx 34.13$$

maximum block pointer = 34 = order of non-leaf node

maximum leaf node = 34

leaf



$x \Rightarrow \text{Key, dp pairs}$

$x (\text{key} + \text{DP}) + BP \leq \text{Block}$

$$x(10+8) + 5 \leq 512$$

$$18x \leq 507$$

$$x \leq \frac{507}{18} \approx 28.16$$

maximum key DP pairs = 28

order of leaf node = 28

transaction को Recover करने के लिए

Deferred Database Modification (Log Based Recovery)

इस file होता है (very small size)
उसमें Action Store करते हैं।

इसको हम लोग

No undo / Redo method
जि करते हैं।

Shared memory

T_1
 $R(A)$
 $A = A + 100$

$W(A)$
 $R(B)$
 $B = B + 200$
 $W(B)$

Commit

Database में Commit होने के बाद में Save कराते हैं

$\langle T_1, \text{Start} \rangle$
 $\langle T_1, A, 200 \rangle$
 $\langle T_1, B, 400 \rangle$
 $\langle T_1, \text{Commit} \rangle$

$\langle T_1, \text{Start} \rangle$
 $\langle T_1, A, 200 \rangle$
 $\langle T_1, B, 400 \rangle$
 $\langle T_1, \text{Commit} \rangle$
 $\langle T_2, \text{Start} \rangle$
 $\langle T_2, C, 500 \rangle$

Redo

new value Store करते हैं।

यदि latest value को save करते हैं

Recovery manager takes the

Decision of Redo

यदि start और commit हुआ है

यदि database नहीं होगा तो new value लिखेंगे। या database होगा तो override करेंगे।

यदि commit नहीं होगा तो Recovery manager Redo operation नहीं perform करेगा तो Roll back करेगा यदि पुरानी value ही database में रखे।

यानि दुरन्त .

Immediate Database Modification (Log Based Recovery)

$A = 100$
 $B = 200$

DB

T_1

$R(A)$

$A = A + 100$

$W(A)$

$R(B)$

$B = B + 200$

$W(B)$

Commit

जैसा यहाँ Ram पर
value 200 हुआ ठीक
उस समय DB में कि
value update हो जायेगा
नहीं

↑
Commit का work
नहीं करते हैं DB
में update करने के लिए

$\langle T_1, \text{Start} \rangle$

$\langle T_1, A, 100, 200 \rangle$

$\langle T_1, B, 200, 400 \rangle$

$\langle T_1, \text{Commit} \rangle$

$\langle T_1, \text{Start} \rangle$

$\langle T_1, A, 1000, 2000 \rangle$

$\langle T_1, B, 5000, 6000 \rangle$

$\langle T_1, \text{Commit} \rangle$

$\langle T_2, \text{Start} \rangle$

$\langle T_2, C, 700, 800 \rangle$

यदि T_1 transaction फाँट हुआ हो तो

Recovery manager ये देखेगा कि
 T_1 इन्फो commit होनी हुआ
है तो redo करेगा

या

यदि commit नहीं हुआ है और transaction
फाँट हो गया है तो वह Undo कर
देगा.

Undo का मतलब पुराना value save कर दे.

यानि

$A = 100$, हो जायेगा.
 $B = 200$

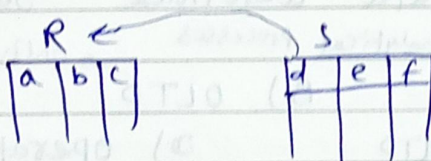
Qus) Let $R(a, b, c)$ and $S(d, e, f)$ be two relations. 'd' is foreign key of S that refers to primary key of R . Consider four operations on ' R ' and ' S '

- (i) insert into R ii) insert into S
 (iii) delete from R iv) delete from S

Which of these can violate referential integrity?

- a) i and ii b) i and iii ✓ c) ii and iii d) i and iv

Soln.



Qus) The view of ~~table~~ total database content is

- ✓ A) Conceptual view B) Internal view
 C) External view d) Physical view

Internal view and External view → Partial view for user.
 Physical view → Storage level use करते हैं,

Qus) In relational mode, Cardinality is ^{degree of attribute}

- ✓ A) No. of tuples B) No. of attributes
 C) No. of tables d) No. of constraints

Ques) _____ Constraint is used to maintain consistency among tuples in two relations

- A) key B) domain ☒ referential integrity
 d) entity integrity.

Big Data (No SQL)

Q.1 what do data Warehouse support ?

- ☒ A) OLAP B) OLTP
C) OLAP and OLTP D) Operational Database
- Online Analytical Processes
→ Online transaction processing.

Q.2 Hadoop is framework that works with variety of related tools. Common group include _____

- ☒ A) Map reduce, Hive and Hbase
 B) Map Reduce, MySQL and Google apps
 C) Map Reduce, Humber, Ignara
 D) Map Reduce, Heron, trumpet.

Q.3 All of following accurately describe Hadoop, Except

- A) Open Source ☒ B) Real time C) Java based.
 d) Distributed Computing approach
- Spark use scala
Real time.

Data model

Data model define how the logical structure of a database is modeled. Data models are fundamental entities to introduce abstraction in DBMS. Data models define how data is connected to each other and how they are processed and stored inside the system.

ER model is best used for the conceptual design of a database

ER model is based on

- Entities and their attributes
 - Relationships among entities.
- { Only two thing in ER model }

the logical association among entities

→ real-world object having properties called attributes
Every attribute is defined by its sets of values called domain.

Strong entity :- at least one attribute which is unique

mysql Command to create table.

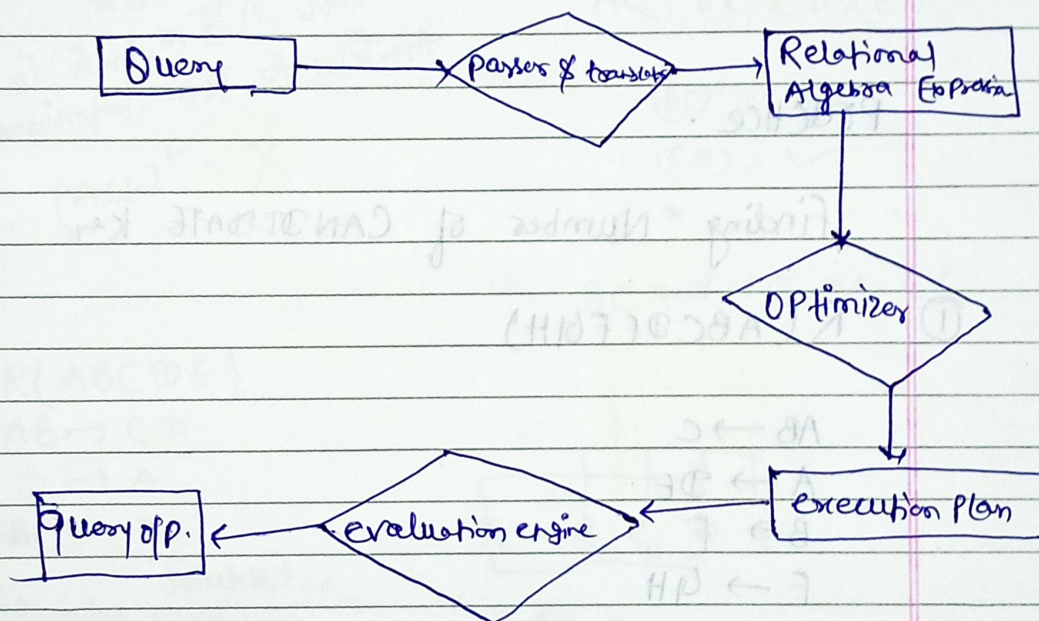
```
CREATE TABLE `testdb`.`alien` (  
  `id` INT NOT NULL,  
  `points` INT NULL,  
  `name` VARCHAR(255) NULL,  
  PRIMARY KEY(`id`));
```


Query Optimization.

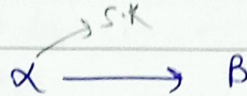
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Basic Steps in Query Optimization.

1. Parsing & translation
2. Optimization.
3. Evaluation

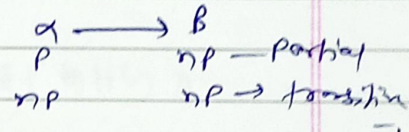


(1) BCNF



if α is S.K then table is in BCNF irrespective of B.

(2) ~~उत्तर~~

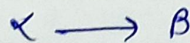


पहले α चेक करेंगे कि S.K है या नहीं. अगर S.K है तो BCNF तब वह 3NF में भी होगा.

अगर α S.K नहीं है तो चेक करेंगे कि B prime attribute है या नहीं. अगर B prime attribute है तो table 3NF में होगा.

OR
(α is C.K or S.K) $\rightarrow$ पहले चेक करेंगे.

(3) 2NF



इसे चेक करने के लिए कोई shortcut नहीं है।

Note :- उपर दो सारे Dependency के लिए चेक करना है।

ये शर्त को Group By Selected value पर चला है न कि
table

Page no: / /

Date: / / Page no: /

Select :- is used to select data from a database.

From :- It is used to list the tables

Where :- It is used to filter the records

Group By :- It can be used in a Select Statement
to collect data across multiple records and
group the results by one or more columns

HAVING :- It is used in combination with the
Group by clause to restrict the group of
returned rows to only those whose condition
is true.

Order By (Keyword) is used to sort the result-set

MIN(), MAX(), SUM(), AVG(), COUNT() →
all are work on Column.

FJWGHSDO

- * The ON Clause Can be used to join tables on
columns that have different names but compatible
data types
- * The Where Clause Can be used to apply additional
conditions in select statements containing the
ON or the Using Clause

Good Question Collection

Page no: / /

Date: / / Page no: /

Department	Staff
department_id	staff_id
department_name	staff_name
department_block_number	department_id

Select department_id, max(count(staff_id)) from staff
Group by department_id



error देगा क्योंकि ही शब्द
हैं max, min, ये सब आ रहा है
और दो से अधिक department_id
से max ले रहा है।

max, min
लगाये हो रक
दिवा चलेगा.

Q. Write a query to display the block numbers in which the maximum number of departments is located.

table	Select department_block_numbers
Department	from
department_id	Department
department_name	group by department_block_number
department_block_number	having count(*) = (select max(cnt)
	from (select count(*) as cnt from
	department group by department_block_numbers) d);

Q. 187 मिनट 43 सेकंड
मि-4/10/11

Qus Write a query to display the name of the department that has the maximum student count.

Department	Student
department_id	Student_id
department_name	Student_name
department_block_number	City
	department_id

```

select department_name from
department (
    select dept.department_name, count(*)
    as std_count from department dept
    join student std on (dept.department_id =
        std.department_id)
    group by dept.department_name order by std_count desc
) where rownum <= 1;
  
```